

Lecture 5: Solving Linear Equations via Cross-Filling

MATH203 Applied Linear Algebra

Motivation

Today's Goal

Learn to solve: $A\mathbf{x} = \mathbf{b}$ using the cross-filling method.

Why?

- Find all solutions (existence)
- Determine uniqueness
- Find basis for null space (relationships between columns)

Method: Decompose augmented matrix $(A \mid \mathbf{b})$ into rank-one pieces, extract simplified equations, solve backwards.

Complete Example: Solving

$$Ax = b$$

Example Setup

Solve:

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 4 \\ 9 \\ 13 \end{pmatrix}$$

Strategy: Form augmented matrix $(A \mid \mathbf{b})$, use cross-filling to simplify, then solve backwards.

Original Equation System

The matrix equation $A\mathbf{x} = \mathbf{b}$ represents three equations:

$$\text{eq1 : } x_1 + 2x_2 + x_3 = 4$$

$$\text{eq2 : } 2x_1 + 4x_2 + x_3 = 9$$

$$\text{eq3 : } 3x_1 + 6x_2 + 2x_3 = 13$$

Goal: Use cross-filling to transform this into a simpler equivalent system.

Step 1: Form Augmented Matrix

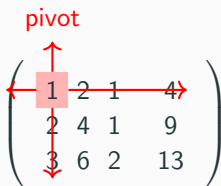
$$(A \mid \mathbf{b})$$
$$\left(\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 2 & 4 & 1 & 9 \\ 3 & 6 & 2 & 13 \end{array} \right)$$

Each row represents one equation from our system.

Next: Choose a pivot to begin cross-filling.

Step 2: Choose First Pivot at (1,1)

pivot

$$\left(\begin{array}{cccc} 1 & 2 & 1 & 4 \\ 2 & 4 & 1 & 9 \\ 3 & 6 & 2 & 13 \end{array} \right)$$


Pivot: entry $(1,1) = 1$ (red)

Cross-filling formula:

$$(R_1 \mid \mathbf{b}_1) = \underbrace{\frac{\text{column 1}}{1}}_{\text{normalized column}} \cdot (\text{row 1})$$

Step 3: First Decomposition Result

$$\underbrace{\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 2 & 4 & 1 & | & 9 \\ 3 & 6 & 2 & | & 13 \end{pmatrix}}_{(A|\mathbf{b})} = \underbrace{\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 2 & 4 & 2 & | & 8 \\ 3 & 6 & 3 & | & 12 \end{pmatrix}}_{(R_1|\mathbf{b}_1) \text{ rank-one}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & | & 1 \\ 0 & 0 & -1 & | & 1 \end{pmatrix}}_{\text{Remainder}}$$

Key observation: Column 1 eliminated from Remainder!

Step 5: Equations from $(R_1 \mid \mathbf{b}_1)$

The matrix:

$$(R_1 \mid \mathbf{b}_1) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 3 & 6 & 3 & 12 \end{array} \right)$$

Each row represents an equation:

$$\text{Row 1 : } 1x_1 + 2x_2 + 1x_3 = 4 \quad (\text{eq1})$$

$$\text{Row 2 : } 2x_1 + 4x_2 + 2x_3 = 8 \quad (2 \cdot \text{eq1})$$

$$\text{Row 3 : } 3x_1 + 6x_2 + 3x_3 = 12 \quad (3 \cdot \text{eq1})$$

Key: All rows = same equation (eq1). Only **one independent equation!**

Step 4: Understanding $(R_1 \mid \mathbf{b}_1)$

How did we get $(R_1 \mid \mathbf{b}_1)$?

$$(R_1 \mid \mathbf{b}_1) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \end{array} \right)$$

This is a **rank-one matrix**: all rows are multiples of row 1.

Step 6: Equations from Remainder

The matrix:

$$\text{Remainder} = \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right)$$

Each row represents an equation:

$$\text{Row 1 : } 0 = 0 \quad (\text{trivial})$$

$$\text{Row 2 : } -x_3 = 1 \quad (\text{this is eq2} - 2 \cdot \text{eq1})$$

$$\text{Row 3 : } -x_3 = 1 \quad (\text{this is eq3} - 3 \cdot \text{eq1})$$

Key: Rows 2 and 3 = same equation ($x_3 = -1$). Only **one independent equation!**

Step 7: Understanding the Decomposition

We decomposed $(A \mid \mathbf{b}) = (R_1 \mid \mathbf{b}_1) + \text{Remainder}$:

Matrix	Equations contained	Relation to original
$(R_1 \mid \mathbf{b}_1)$	eq1 ($\times 3$ rows)	eq1
Remainder	$-x_3 = 1$ ($\times 2$ rows)	eq2 $- 2 \cdot \text{eq1}$, eq3 $- 3 \cdot \text{eq1}$

Simplified equation system:

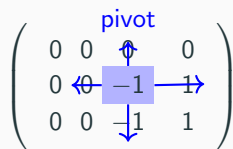
$$\text{From } R_1 : \quad x_1 + 2x_2 + x_3 = 4$$

$$\text{From Remainder : } \quad x_3 = -1$$

This is equivalent to the original three equations!

Step 8: Continue Cross-Filling on Remainder

The Remainder still has structure. Choose second pivot at $(2, 3) = -1$:



The diagram shows a 3x4 matrix with the second row and third column highlighted in light blue. The pivot element at row 2, column 3 is -1. Blue arrows indicate the following operations: a vertical arrow pointing down from the pivot to the element at (3,3) which is -1; a horizontal arrow pointing left from the pivot to the element at (2,2) which is 0; a horizontal arrow pointing right from the pivot to the element at (2,4) which is 1; and a vertical arrow pointing up from the pivot to the element at (1,3) which is 0. The word "pivot" is written in blue above the pivot element.

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

$$(R_2 \mid \mathbf{b}_2) = \frac{\text{column 3 of Remainder}}{-1} \cdot (\text{row 2 of Remainder})$$

Step 9: Second Decomposition Result

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & -1 & | & 1 \\ 0 & 0 & -1 & | & 1 \end{pmatrix}}_{\text{First Remainder}} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & -1 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}}_{(R_2 | \mathbf{b}_2) \text{ rank-one}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}}_{\text{All zeros!}}$$

Cross-filling complete: $(A | \mathbf{b}) = (R_1 | \mathbf{b}_1) + (R_2 | \mathbf{b}_2)$

Step 10: Equations from $(R_2 \mid \mathbf{b}_2)$

The matrix:

$$(R_2 \mid \mathbf{b}_2) = \left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{array} \right)$$

Each row represents an equation:

$$\text{Row 1 : } 0 = 0 \quad (\text{trivial})$$

$$\text{Row 2 : } x_3 = -1 \quad (\text{eq2} - 2 \cdot \text{eq1})$$

$$\text{Row 3 : } x_3 = -1 \quad (\text{eq3} - 3 \cdot \text{eq1})$$

Key: All rows = same equation ($x_3 = -1$). Only **one independent equation**!

Step 11: The Simplified Equation System

From the cross-filling decomposition, we extracted:

From piece	Equation
$(R_1 \mid \mathbf{b}_1)$	$x_1 + 2x_2 + x_3 = 4$ (pivot: x_1 , free: x_2 , x_3 not yet determined)
$(R_2 \mid \mathbf{b}_2)$	$x_3 = -1$ (pivot: x_3)

Variable classification (from cross-filling):

- **Pivot variables:** x_1, x_3 (columns 1 and 3 where we chose pivots)
- **Free variable:** x_2 (column 2, non-pivot column)

Step 12: Backward Solving

Solve from the **simplest equation** backwards:

From equation 2:

$$x_3 = -1$$

Substitute into equation 1:

$$x_1 + 2x_2 + x_3 = 4$$

$$x_1 + 2x_2 + (-1) = 4$$

$$x_1 + 2x_2 = 5$$

$$x_1 = 5 - 2x_2$$

Variable x_2 is not determined by any equation!

Step 13: Identify Pivot and Free Variables

From cross-filling, we chose pivots at:

- Column 1 (first pivot)
- Column 3 (second pivot)

Classification:

- **Pivot variables:** x_1, x_3 (determined by equations)
- **Free variable:** x_2 (can be chosen freely)

Pivot columns = columns where we chose pivots.

Non-pivot columns = remaining columns $\rightarrow$ free variables.

Step 14: General Solution

Let $x_2 = t$ (free parameter).

From our backward solving:

$$x_1 = 5 - 2t$$

$$x_2 = t$$

$$x_3 = -1$$

General solution:

$$\mathbf{x} = \begin{pmatrix} 5 - 2t \\ t \\ -1 \end{pmatrix}$$

This describes **infinitely many solutions** (one for each value of t).

Step 15: Decompose into Particular + Null Space

Rewrite the general solution:

$$\mathbf{x} = \begin{pmatrix} 5 - 2t \\ t \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

Two parts:

- **Particular solution** $\mathbf{x}_p = \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix}$ (when $t = 0$)
- **Null space direction** $\mathbf{v} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ (coefficient of t)

Check: $A\mathbf{v} = \mathbf{0}$ (verify this is in null space).

Step 16: Verification (Particular Solution)

Verify particular solution: $A\mathbf{x}_p = \mathbf{b}$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 + 0 - 1 \\ 10 + 0 - 1 \\ 15 + 0 - 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \\ 13 \end{pmatrix}$$

✓

Step 16: Verification (Null Space Vector) (continued)

Verify null space vector: $A\mathbf{v} = \mathbf{0}$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 + 2 + 0 \\ -4 + 4 + 0 \\ -6 + 6 + 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

✓

Both parts of the solution are verified!

Summary of the Example

What we did:

1. Formed augmented matrix ($A \mid \mathbf{b}$)
2. Used cross-filling to decompose into rank-one pieces
3. Extracted simplified equations from each piece
4. Solved backwards from simplest equation
5. Identified free variables
6. Wrote general solution = particular + null space

Key insight: Cross-filling transforms the original equation system into an **equivalent simpler system** where variables decrease step-by-step.

Why Variables Decrease

Observation: Variables Decrease with Each Piece

In our example:

Piece	Active variables	Count
Original ($A \mid \mathbf{b}$)	x_1, x_2, x_3	3
After extracting R_1 Remainder	x_3 only	1
After extracting R_2 Final Remainder	none	0

Pattern: Each cross-filling step eliminates at least one variable from the remainder.

Why Does This Happen?

Cross-filling structure:

$$(R_i \mid \mathbf{b}_i) = \frac{\text{column } j}{a_{kj}} \cdot (\text{row } k)$$

- All rows of $(R_i \mid \mathbf{b}_i)$ are multiples of row k
- When we subtract $(R_i \mid \mathbf{b}_i)$ from the matrix, column j becomes zero in the remainder
- Column j is **eliminated** from future consideration

Result: Each step reduces the number of active columns (variables) by at least one.

Consequence: Backward Solving Works

Since variables decrease:

- Last piece ($R_r \mid \mathbf{b}_r$) has **fewest variables**
- Second-to-last has one more variable
- $\vdots$
- First piece has most variables

Backward solving strategy:

1. Start from the last equation (fewest variables)
2. Solve for pivot variable(s)
3. Substitute into previous equation
4. Repeat until all pivot variables determined

Existence and Uniqueness

When Does $Ax = b$ Have a Solution?

Cross-filling $(A \mid \mathbf{b})$ can produce two outcomes:

Case 1: All remainder rows eventually become zero.

- No contradiction
- Solution **exists**
- Example: our 3×3 example

Case 2: Some remainder row becomes $\left(0 \quad 0 \quad \cdots \quad 0 \mid c \right)$ where $c \neq 0$.

- This represents equation $0 = c$ (contradiction!)
- No solution exists

Theorem (Existence)

$A\mathbf{x} = \mathbf{b}$ has a solution if and only if:

$$\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$$

Interpretation:

- If adding $\mathbf{b}$ as a column doesn't increase rank $\Rightarrow \mathbf{b}$ is already in the span of A 's columns
- Equivalently: $\mathbf{b} \in \text{Col}(A)$

Uniqueness Criterion

If a solution exists, is it unique?

Theorem (Uniqueness)

If $A\mathbf{x} = \mathbf{b}$ has a solution, it is **unique** if and only if:

$$\text{Null}(A) = \{\mathbf{0}\}$$

Equivalently:

- No free variables
- $\text{rank}(A) = n$ (number of columns)

Proof idea: If $\mathbf{x}_1, \mathbf{x}_2$ both solve $A\mathbf{x} = \mathbf{b}$, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.

The Four Cases

	$\text{rank}(A) = n$ (no free vars)	$\text{rank}(A) < n$ (has free vars)
$\text{rank}(A \mid \mathbf{b})$ $= \text{rank}(A)$ (consistent)	Unique solution	Infinitely many solutions
$\text{rank}(A \mid \mathbf{b})$ $= \text{rank}(A) + 1$ (inconsistent)	No solution	No solution

Our example: $\text{rank}(A) = 2$, $\text{rank}(A \mid \mathbf{b}) = 2$, $n = 3 \Rightarrow$ infinitely many solutions.

Finding Null Space Basis

Definition: Null Space

Definition (Null Space)

For a matrix $A \in \mathbb{R}^{m \times n}$:

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Meaning: Vectors that get mapped to zero.

Interpretation: Relationships between the columns of A .

If $\mathbf{v} \in \text{Null}(A)$, then $v_1\mathbf{a}_1 + \cdots + v_n\mathbf{a}_n = \mathbf{0}$ (linear dependence).

Example: Finding Null Space Basis for Our Matrix

Solve $A\mathbf{x} = \mathbf{0}$:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}$$

Form $(A \mid \mathbf{0})$ and cross-fill (same steps as before, but with $\mathbf{b} = \mathbf{0}$):

Simplified system:

$$x_1 + 2x_2 + x_3 = 0$$

$$x_3 = 0$$

Backward Solving for Null Space

From the simplified system:

$$x_3 = 0$$

$$x_1 + 2x_2 + 0 = 0 \quad \Rightarrow \quad x_1 = -2x_2$$

Free variable: $x_2 = t$

General null space vector:

$$\mathbf{x} = \begin{pmatrix} -2t \\ t \\ 0 \end{pmatrix} = t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

Basis for Null(A): $\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \right\}$

General Procedure: Finding Null Space Basis

Proposition

To find a basis for $\text{Null}(A)$:

1. Solve $A\mathbf{x} = \mathbf{0}$ using cross-filling
2. Identify free variables (non-pivot columns)
3. For each free variable: set it to 1, others to 0, solve for pivots
4. Each setting gives one basis vector

Dimension of Null (A) = number of free variables.

Rank-Nullity Theorem

Counting Pivot and Free Variables

For an $m \times n$ matrix A :

- Total columns: n
- Pivot columns (from cross-filling): $r = \text{rank}(A)$
- Non-pivot columns (free variables): $n - r$

But non-pivot columns = dimension of null space!

$$\dim(\text{Null}(A)) = n - r$$

Rank-Nullity Theorem

Theorem (Rank-Nullity Theorem)

For any $m \times n$ matrix A :

$$\text{rank}(A) + \dim(\text{Null}(A)) = n$$

Proof: Every column is either:

- A pivot column (contributes to rank), or
- A non-pivot column (contributes one basis vector to null space)

These counts partition the n columns.

Also written: $\text{rank}(A) + \text{nullity}(A) = n$

Example Verification

Our 3×3 example:

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}$$

- $n = 3$ (three columns)
- $\text{rank}(A) = 2$ (two pivot columns: 1 and 3)
- $\dim(\text{Null}(A)) = 1$ (one free variable: x_2)

Check: $2 + 1 = 3$ ✓

Summary

What We Learned

Cross-filling method for solving $Ax = b$:

1. Decompose $(A \mid \mathbf{b})$ into rank-one pieces
2. Each piece gives one simplified equation
3. Variables decrease with each piece
4. Solve backwards from simplest equation

Key results:

- Existence: $\text{rank}(A \mid \mathbf{b}) = \text{rank}(A)$
- Uniqueness: $\text{Null}(A) = \{\mathbf{0}\}$
- General solution = particular + null space
- Rank-Nullity: $\text{rank}(A) + \text{nullity} = n$

Lecture 6: Four Fundamental Subspaces

We'll explore:

- Column space and null space (from today)
- Row space and left null space
- Orthogonality relationships: $\text{Null}(A) \perp \text{Row}(A)$
- The fundamental theorem of linear algebra